Mathematical Model: Agent Political Simulation

(generated from codebase)

1 Scope and State

This document models the discrete-time simulation implemented in:

- `src/simulation/engine.py` (turn loop)
- `src/actions/decision.py` (action choice)
- `src/actions/voting.py` (voting and veto)
- `src/simulation/consequences.py` (state updates)
- `src/simulation/drift.py` (ideology drift and IG radicalization)
- `src/simulation/elections.py`, `src/simulation/factions.py` (elections/splits)

1.1 Sets and indices

Let:

$t \in \{0, 1, 2, \dots\}$	turn index
$\mathcal{P}$	set of places (jurisdictions)
$\mathcal{K}$	set of parties
$\mathcal{G}$	set of interest groups
$\mathcal{A}_p(t) \subseteq \mathcal{A}(t)$	agents currently in place p
$\mathcal{L}_p(t) \subseteq \mathcal{A}_p(t)$	legislators in place p
$\mathcal{X}$	set of ideology axes (e. g. economic, social)

1.2 World state variables

For each agent $a \in \mathcal{A}(t)$:

$\mathbf{i}_a(t) \in [-1, 1]^{\lvert\mathcal{X}\rvert}$	ideology vector
$k(a, t) \in \mathcal{K}$	party membership
$p(a, t) \in \mathcal{P}$	place membership
$\alpha_{a,g}(t) \in [0, 1]$	allegiance weight to IG g
$r_{a,b}(t) \in [-1, 1]$	relationship/trust (directed)
$\pi_{a,g}(t) \in [0, 1]$	popularity with IG g
$s_a(t) \in [0, 1]$	party standing (loyalty)
$\eta_a(t) \in [0, 1]$	ambition (noise scale)
$\mathrm{arch}(a) \in \{\text{LOYALIST, OPPORTUNIST, IDEOLOGUE}\}$	voting archetype

For each interest group $g \in \mathcal{G}$ and place $p \in \mathcal{P}$:

$$\begin{aligned}
\sigma_{g,p}(t) &\in [0,1] && \text{satisfaction} \\
e_{g,p}(t) &\in [0,1] && \text{electorate share (mobilized strength)} \\
\mathbf{i}_g(t) &\in [-1,1]^{|\mathcal{X}|} && \text{IG ideology vector}
\end{aligned}$$

For each party $k \in \mathcal{K}$:

$$\begin{aligned}
\mathbf{i}_k(t) &\in [-1,1]^{|\mathcal{X}|} && \text{party ideology vector} \\
\rho_{k,k'}(t) &\in [-1,1] && \text{party-to-party relation} \\
\rho^{IG}_{k,g}(t) &\in [0,1] && \text{party-to-IG relation} \\
\theta_k &\in [0,1] && \text{directive threshold (party-line activation)} \\
B_k(t) &\in \mathbb{Z}_{\geq 0} && \text{campaign budget (elections)}
\end{aligned}$$

1.3 Policy objects (ephemeral)

During each turn, each government generates a pool of candidate policies $\mathcal{M}_p(t)$ (not persisted in `World`; see `src/models/policy.py`). For each policy m:

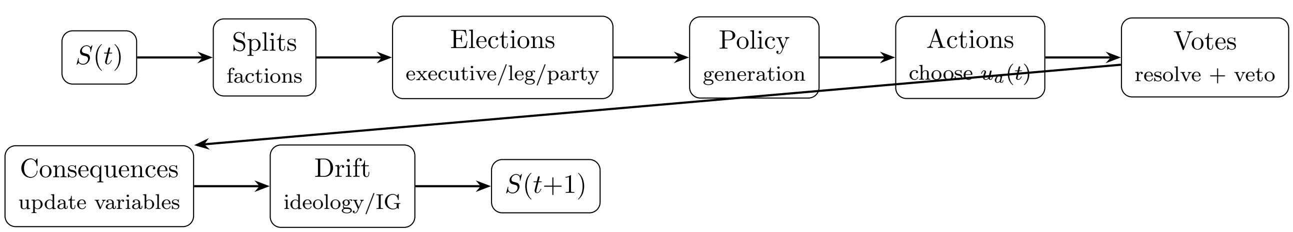
$$\begin{aligned}
\Delta_{m,g} &\in [-1,1] && \text{impact on IG } g \\
\mathbf{d}_m &\in \mathbb{R}^{|\mathcal{X}|} && \text{ideology alignment directions by axis}
\end{aligned}$$

2 Turn structure as a stochastic state transition

The simulation is a discrete-time stochastic dynamical system:

$$S(t+1) = F(S(t), \xi_t),$$

where $S(t)$ is the full world state and ξ_t is exogenous randomness (the RNG draws).



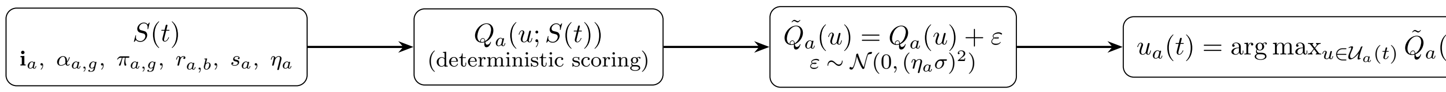
In code (`src/simulation/engine.py`), a turn is:

1. splits and elections (may change roster/roles)
2. per-place policy generation
3. per-agent action selection
4. legislative votes + executive veto + possible override
5. apply consequences of non-vote actions
6. ideological drift and IG radicalization/moderation

3 Action selection model

3.1 Action set

For each agent a, an availability filter defines a feasible set of actions $\mathcal{U}_a(t)$ (see `src/actions/availability.py`). The decision policy scores candidates and selects the maximum after adding ambition-scaled Gaussian noise (`src/actions/decision.py`).



3.2 Utility weights

Each agent has a weight vector

$$\mathbf{w}_a = (w_a^{pop}, w_a^{stand}, w_a^{ideo}, w_a^{ig}, w_a^{rel}), \quad \sum_i w_i = 1,$$

defaulting to `DEFAULT_WEIGHTS` in `src/actions/utility.py`.

3.3 Noisy argmax decision rule

For each candidate action $u \in \mathcal{U}_a(t)$ the code forms a deterministic score $Q_a(u; S(t))$, then adds noise:

$$\tilde{Q}_a(u) = Q_a(u; S(t)) + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, (\eta_a(t)\sigma_u)^2),$$

with $\sigma_u \in \{0.02, 0.01\}$ depending on (agent vs party-leader) action. Then:

$$u_a(t) = \arg\max_{u \in \mathcal{U}_a(t)} \tilde{Q}_a(u).$$

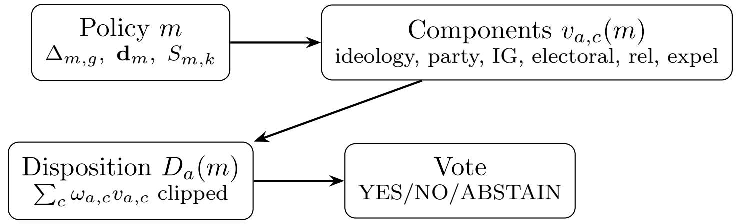
4 Voting model

4.1 Vote disposition

When a policy m is put to a vote, each legislator $a \in \mathcal{L}_p(t)$ computes a disposition score $D_a(m) \in [-1, 1]$ (see `compute_vote_disposition` in `src/actions/voting.py`):

$$D_a(m) = \mathrm{clip}_{[-1,1]}\left(\sum_{c \in \mathcal{C}} \omega_{a,c}\, v_{a,c}(m)\right),$$

where components $\mathcal{C} = \{$ideology, party_directive, ig_pressure, electoral, relationships, expulsion_risk$\}$, with archetype-conditioned weights $\omega_{a,c}$ (possibly perturbed once per agent and stored).



4.2 Component definitions (code-faithful)

Ideology alignment. If $\mathbf{d}_m$ is the policy's axis-direction map and $\mathbf{i}_a$ agent ideology:

$$v_{a,\text{ideo}}(m) = \frac{1}{|\mathcal{X}_m|} \sum_{x \in \mathcal{X}_m} i_{a,x}\, d_{m,x}.$$

Party directive pressure. Policies may have an explicit party support scalar $S_{m,k} \in [-1, 1]$ stored as `policy.attributes["party_support"][party]`. Given party threshold θ_k:

$$v_{a,\text{party}}(m) = \begin{cases} \frac{S_{m,k}-\theta_k}{1-\theta_k} & S_{m,k} > \theta_k \\ \frac{S_{m,k}+\theta_k}{1-\theta_k} & S_{m,k} < -\theta_k \\ 0 & \text{otherwise.} \end{cases}$$

IG pressure. Let $\Delta_{m,g}$ be impact and $\alpha_{a,g}$ allegiance. The code dampens low allegiance:

$$\alpha'_{a,g} = \begin{cases} \alpha_{a,g} & \alpha_{a,g} \geq 0.15 \\ 0.4\alpha_{a,g} & \alpha_{a,g} < 0.15 \end{cases}$$

and then

$$v_{a,\text{ig}}(m) = \frac{\sum_g \Delta_{m,g}\alpha'_{a,g}}{\sum_g \alpha'_{a,g}} \quad \text{(0 if denominator is 0)}.$$

Electoral. Popularity-weighted impact:

$$v_{a,\text{elect}}(m) = \frac{\sum_g \Delta_{m,g}\pi_{a,g}}{\sum_g \pi_{a,g}} \quad \text{(0 if denominator is 0)}.$$

Relationships.

$$v_{a,\text{rel}}(m) = r_{a,\text{proposer}(m)}.$$

Expulsion risk. If directive pressure is $v_{a,\text{party}}(m) \neq 0$, vulnerability in the code is:

$$\nu_a = 0.45(1 - s_a) + 0.35(1 - \bar{\pi}_a) + 0.20(1 - \text{altAccess}_a),$$

clipped to $[0, 1]$, and then

$$v_{a,\text{expel}}(m) = v_{a,\text{party}}(m) \cdot \nu_a.$$

(Here $\bar{\pi}_a$ is electorate-share-weighted popularity survivability, and altAccess_a depends on ideological distance to other parties as in code.)

4.3 Vote resolution, veto, and override

Each legislator casts:

$$\text{YES if } D_a(m) > 0.1, \quad \text{NO if } D_a(m) < -0.1,$$

else a gray-zone rule that can abstain when personal lean is negative but party pull is positive.

Define indicator functions for each legislator $a \in \mathcal{L}_p(t)$:

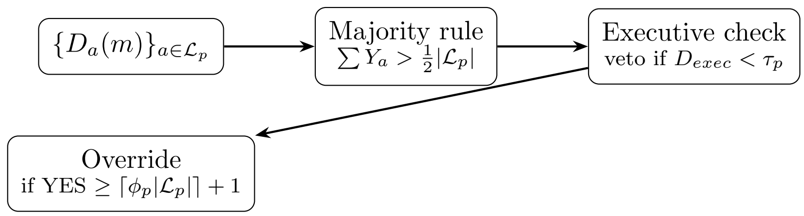
$$Y_a(m) = \mathbb{I}[D_a(m) > 0.1], \qquad N_a(m) = \mathbb{I}[D_a(m) < -0.1].$$

The bill passes if:

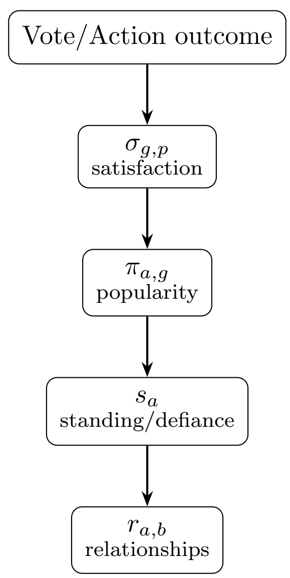
$$\sum_{a \in \mathcal{L}_p(t)} Y_a(m) > \frac{1}{2} |\mathcal{L}_p(t)|.$$

If tied (YES=NO) and an executive exists, executive breaks tie in favor iff their $D_{exec}(m) > 0$.

If passed, executive vetoes iff $D_{exec}(m) < \tau_p$ where τ_p is `gov.veto_threshold` (default -0.3). Veto override is checked later using a fraction ϕ_p (default 2/3).



5 Consequences (state updates)



5.1 Interest-group satisfaction update

On passage, for each impacted group with satisfaction defined in the place:

$$\sigma_{g,p} \leftarrow \text{clip}_{[0,1]} \left(\sigma_{g,p} + \Delta_{m,g} \cdot c(\Delta_{m,g}) \right),$$

where in code $c = 0.5$ for $\Delta \geq 0$ and $c = 0.6$ for $\Delta < 0$ (negatives hit harder).

There is also a cross-IG conflict penalty for pairs of groups with high ideological conflict.

5.2 Agent popularity update (memory of vote)

For each voter a and IG g:

$$\pi_{a,g} \leftarrow \text{clip}_{[0,1]} \left(\pi_{a,g} + \text{dir}(a)\, \Delta_{m,g} \cdot 0.18 \right),$$

where $\text{dir}(a) = +1$ if voted YES else -1.

5.3 Party standing update

If a party directive is active, following it increases standing; defying decreases it:

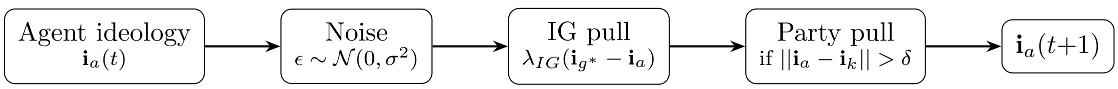
$$s_a \leftarrow \mathrm{clip}_{[0,1]}(s_a \pm (0.01 + 0.03|v_{a,\mathrm{party}}|) \text{ or } (0.02 + 0.08|v_{a,\mathrm{party}}|)).$$

The code also tracks `recent_defiance`.

5.4 Relationship update

Voting together increases mutual trust (+0.02); opposing votes decrease trust (-0.01). Non-vote actions also adjust relationships and standing (see `src/simulation/consequences.py`).

6 Ideological drift and IG radicalization



6.1 Agent ideology drift

For each low-detail-level agent and axis x (see `src/simulation/drift.py`):

$$i_{a,x} \leftarrow \mathrm{clip}_{[-1,1]}\Big(i_{a,x} + \epsilon_{a,x} + \lambda_{IG}(i_{g^*,x} - i_{a,x}) + \nVdash[|i_{a,x} - i_{k,x}| > \delta]\lambda_{party}(i_{k,x} - i_{a,x})\Big),$$

with $\epsilon_{a,x} \sim \mathcal{N}(0, \sigma^2)$.

6.2 IG radicalization / moderation

Let $\bar{\sigma}_g$ be average satisfaction across places. If $\bar{\sigma}_g < \sigma_{rad}$, the IG becomes more extreme and more mobilized:

$$i_{g,x} \leftarrow \mathrm{clip}(i_{g,x} \cdot \gamma + \mathrm{sign}(i_{g,x}) \cdot 0.005), \qquad e_{g,p} \leftarrow \min(1, e_{g,p} \cdot \kappa).$$

If $\bar{\sigma}_g > \sigma_{cent}$, it moderates and slightly demobilizes.

7 Tunable Variables Table (code-extracted)

The following are the primary *tweak points* exposed as fields or constants in the code. Some are hard-coded constants; others live in dataclasses and can be changed at initialization time.

Variable	**Where**	**Default**	**Effect**
t	`models/world.py`	0	Turn counter.
$\mathbf{i}_a$	`models/agent.py`	—	Ideological position per axis in $[-1, 1]$.
$\alpha_{a,g}$	`models/agent.py`	{}	IG influence weights on agent actions/votes.
$\pi_{a,g}$	`models/agent.py`	{}	IG-specific approval that affects elections/vulnerability.

Variable	Where	Default	Effect
$r_{a,b}$	`models/agent.py`	{}	Inter-agent trust used in votes and actions.
s_a	`models/agent.py`	0.5	Loyalty/standing inside party; used in voting/expulsion risk.
η_a	`models/agent.py`	0.5	Noise scale for action choice; affects randomness.
$\mathbf{w}_a$	`actions/utility.py`	`DEFAULT_WEIGHTS`	Per-agent action utility coefficients.
$\omega_{a,c}$	`actions/voting.py`	(generated)	Per-agent vote component coefficients.
θ_k	`models/party.py`	0.15	Party-line activation threshold for directives.
B_k	`models/party.py`	3	Election nomination budget decrementing on use.
—	`models/party.py`	0.3	Nomination threshold.
—	`models/party.py`	20	Leadership election interval.
τ_p	`models/government.py`	-0.3	Executive veto if exec disposition below threshold.
ϕ_p	`models/government.py`	2/3	Fraction of seats needed (plus 1) to override veto.
—	`simulation/elections.py`	—	Per-government election schedule interval.
—	`actions/utility.py`	(0.3,0.2,0.15,0.2,0.15)	Default utility weights `DEFAULT_WEIGHTS`.
—	`actions/voting.py`	varies	Archetype vote weights `ARCHETYPE_WEIGHTS`.
—	`actions/voting.py`	±0.1	Vote thresholds for YES/NO/abstain logic.
σ	`simulation/drift.py`	0.01	Std-dev of per-turn ideology Brownian noise.
λ_{IG}	`simulation/drift.py`	0.02	Pull rate toward top-allegiance IG ideology.
δ	`simulation/drift.py`	0.4	Divergence threshold before party-pull activates.
λ_{party}	`simulation/drift.py`	0.015	Pull rate back toward party ideology.
σ_{rad}	`simulation/drift.py`	0.25	Avg satisfaction below triggers IG radicalization.
γ	`simulation/drift.py`	1.05	Per-turn multiplier pushing IG ideology away from zero.
κ	`simulation/drift.py`	1.02	Per-turn multiplier increasing IG electorate share.
σ_{cent}	`simulation/drift.py`	0.72	Avg satisfaction above triggers IG moderation.
—	`simulation/drift.py`	0.03	Centrist pull rate.
—	`simulation/drift.py`	0.99	Centrist share decay.
—	`simulation/elections.py`	2	Executive term limit.
—	`simulation/elections.py`	4	Legislative term limit.

Variable	Where	Default	Effect
—	`simulation/elections.py`	10	Stale-agent removal threshold.
—	`simulation/elections.py`	0.25	Support dilution factor.
—	`simulation/elections.py`	0.6	Viability ratio.
—	`simulation/factions.py`	0.45	Split satisfaction threshold.
—	`simulation/factions.py`	3	Split inertia.
—	`simulation/factions.py`	0.10	Split ideology gap.
—	`simulation/factions.py`	2	Minimum splinter size.
—	`simulation/factions.py`	2	Splinter start budget.
—	`simulation/factions.py`	0.5	Ambition floor.

Math ↔ code mapping

Math variable	Code	Notes
t	`World.turn`	Turn counter.
$\mathbf{i}_a$	`Agent.ideology.axes`	Ideology by axis name.
$\mathbf{i}_k$	`Party.ideology.axes`	Party ideology by axis name.
$\mathbf{i}_g$	`InterestGroup.ideology.axes`	IG ideology by axis name.
$\alpha_{a,g}$	`Agent.allegiances[InterestGroup]`	Weighted allegiances.
$\pi_{a,g}$	`Agent.popularity[InterestGroup]`	Popularity per IG.
$r_{a,b}$	`Agent.relationships[Agent]`	Directed trust edges.
s_a	`Agent.party_standing`	Party loyalty/standing.
η_a	`Agent.ambition`	Action-selection noise scaling.
$\mathbf{w}_a$	`Agent.attributes["utility_weights"]`	Defaults to `DEFAULT_WEIGHTS`.
$\omega_{a,c}$	`Agent.attributes["vote_weights"]`	Seeded from `ARCHETYPE_WEIGHTS`.
$\sigma_{g,p}$	`InterestGroup.satisfaction[Place]`	Satisfaction per place.
$e_{g,p}$	`InterestGroup.electorate_share[Place]`	Electorate share per place.
θ_k	`Party.directive_threshold`	Directive activation threshold.
B_k	`Party.campaign_budget`	Used during nominations.
τ_p	`Government.veto_threshold`	Executive veto cutoff.
ϕ_p	`Government.veto_override_fraction`	Override fraction in `engine.py`.
σ	`_RANDOM_DRIFT_SIGMA`	Drift noise in `simulation/drift.py`.
λ_{IG}	`_IG_PULL_RATE`	Drift IG pull.
δ	`_PARTY_PULL_THRESHOLD`	Party pull threshold.
λ_{party}	`_PARTY_PULL_RATE`	Party pull rate.
σ_{rad}	`_RAD_THRESHOLD`	IG radicalization threshold.
γ	`_RAD_AXIS_SCALE`	IG radicalization multiplier.
κ	`_RAD_SHARE_BOOST`	IG mobilization multiplier.
σ_{cent}	`_CENTRIST_THRESHOLD`	IG moderation threshold.